ELSEVIER

Journal of South American Earth Sciences 25 (2008) 86–99

Journal of
South American
Earth Sciences

www.elsevier.com/locate/jsames

Paleogeographic and tectonic implications of the first paleomagnetic results from the Middle–Late Cambrian Mesón Group: NW Argentina

Cecilia M. Spagnuolo [a,*], Augusto E. Rapalini [a], Ricardo A. Astini [b]

[a] *INGEODAV, Depto.Cs.Geológicas, F.C.E.yN., Universidad de Buenos Aires, Pabellón 2, Ciudad Universitaria, C1428EHA, Buenos Aires, Argentina*
[b] *Laboratorio de Análisis de Cuencas, Cátedra de Estratigrafía y Geología Histórica, F.C.E.F.yN., Universidad Nacional de Córdoba, Av. Velez Sarsfield 1611, X5016GCA, Córdoba, Argentina*

Abstract

The first paleomagnetic data from autochthonous Cambrian rocks in NW Argentina is reported to constrain the apparent polar wander path (APWP) of Gondwana during the Early Paleozoic. The paleomagnetic pole (Lat 4.5°S; Long 359.0°E; dp = 5.5; dm = 8.8; $n = 26$) was obtained from the red to purple sandstones of the Campanario Formation. These rocks present a characteristic remanence carried by fine-grained hematite. The pole indicates that during the Late Cambrian, rocks now exposed in the Eastern Cordillera of NW Argentina were deposited at relatively low latitudes (≈26°S), consistent with intense chemical alteration during wet and warm climates, as indicated from petrography of apparent first-cycle quarzites of the Mesón Group. However, the pole position is anomalous with respect to the most accepted apparent polar wander paths for Gondwana, suggesting that the study area (22°50′S, 65°00′W) underwent clockwise rotation of 38° ± 8°, likely related to the Cenozoic central Andes rotation pattern characteristic of the region.
© 2007 Elsevier Ltd. All rights reserved.

Keywords: Gondwana; Paleomagnetism; Cambrian; Orocline; Northwestern Argentina

Resumen

Se obtuvo el primer polo paleomagnético Cámbrico del NO Argentino para tratar de determinar con mayor presición la curva de deriva polar aparente (CDPA) del Gondwana durante el Paleozoico Temprano. El polo paleomagnético (Lat = 4,5°S; Long = 359,0°E; dp = 5,5; dm = 8,8; $n = 26$) fue obtenido de las areniscas rojas a moradas de la Formación Campanario. Estas rocas presentan una remanencia característica portada por hematita de grano fino. El polo indica que durante el Cámbrico tardío las rocas expuestas actualmente en la Cordillera Oriental de Argentina se depositaron a latitudes relativamente bajas (≈26°S). Este resultado es consistente con una intensa alteración química de las cuarcitas de primer ciclo del Grupo Mesón durante climas húmedos y cálidos según lo indican estudios petrográficos. Sin embargo, la posición polar obtenida es anómala respecto a la curva de deriva polar aparente del Gondwana más aceptada hasta el momento, sugiriendo que la zona de estudio (22°50′S, 65°00′O) sufrió una rotación horaria de 38° ± 8°, relacionada probablemente al patrón cenozoico de rotaciones de los Andes Centrales característico de la región.
© 2007 Elsevier Ltd. All rights reserved.

Palabras clave: Gondwana; Paleomagnetismo; Cámbrico; Oroclino; Noroeste argentino

* Corresponding author.
E-mail addresses: cspagnuolo@gl.fcen.uba.ar (C.M. Spagnuolo), rapalini@gl.fcen.uba.ar (A.E. Rapalini), raastini@efn.uncor.edu (R.A. Astini).

0895-9811/$ - see front matter © 2007 Elsevier Ltd. All rights reserved.
doi:10.1016/j.jsames.2007.08.004

1. Introduction

The Andean chain changes strike from NW–SE to N–S at approximately 18°S latitude. This region is generally known as the "Bolivian Orocline" and comprises part of Bolivia, Perú, and northern Chile and Argentina. It is characterized by counterclockwise rotations north of the bend and clockwise rotations south of it (Somoza et al., 1996; Beck, 1998; Randall, 1998).

Because of the lack of Early Paleozoic paleomagnetic data from northwestern Argentina, a reconnaissance paleomagnetic survey was carried out on the Middle–Late Cambrian Mesón Group, exposed along the Iruya River in the province of Salta (22°49′ – 22°50′S; 64°50′ – 65°00′W), near the Matancillas anticline (Fig. 1). The results of this study permit us to compute for the first time a Cambrian paleomagnetic pole for areas of Argentina considered autochthonous. However, the position of the computed pole suggests that the area underwent a significant clockwise rotation, presumably related to the rotation pattern associated with the Bolivian Orocline.

2. Geological setting and stratigraphy

The study area is located in the eastern limit of the Eastern Cordillera (Cordillera Oriental) along the Iruya River in the Salta province of Argentina (Fig. 1).

This region belongs to the Central Andes (Gansser, 1973; Ramos, 1999a) and is affected by normal subduction. It is characterized by an active volcanic arc in the Western Cordillera (Cordillera Occidental), thermal uplift, maximum crustal thickening, and the largest crustal shortening within the Andean Cordillera (Allmendinger et al., 1997). The Eastern Cordillera is limited to the west by the Puna plateau and to the east by the sub-Andean ranges (Sierras Subandinas) (Mon, 1999; Ramos, 1999b); the boundary between them is tectonic (Mon and Salfity, 1995). The structure of the Eastern Cordillera in the Andean backarc region is controlled by normal subduction (~30°) of the Nazca plate (Ramos et al., 2004). The regional structures of the Eastern Cordillera consist of a dominant, east-vergent, thick-skinned thrust system with a Proterozoic basement composed of low-grade metamorphic greywackes of the Puncoviscana Formation. The thrusts stack has produced a high topographic relief in the region known as the Interandean belt (Kley, 1996).

Many authors (Hernández et al., 1996; Ramos et al., 2004) propose that the main thrust sequence that uplifted the Eastern Cordillera and sub-Andean ranges occurred between 17 and 12 Ma, with Pliocene and Quaternary pulses of deformation.

The stratigraphy of the region consists of the Puncoviscana Formation and the Mesón and Santa Victoria groups. The Puncoviscana Formation (Turner, 1960), composed of metasediments, is exposed along N–S-trending outcrops. It has been assigned to the Ediacaran–Early Cambrian because of its relative stratigraphic position (Turner, 1960), the presence of ichnofossils (Buatois and Mangano, 2003), K–Ar analyses of fine-grained fractions (Do Campo and Guevara Ribeiro, 2005), the age of intrusive bodies (Turner, 1960), detrital zircons, and the age of regional metamorphism (see Omarini et al., 1999; Ramos, 2000; Schwartz and Gromet, 2004).

The Mesón Group (Turner, 1960) crops out extensively in northwestern Argentina. Its age is bracketed between the Middle and Late Cambrian, according to stratigraphic considerations (Turner, 1960), and between the Lower and Middle Cambrian according to ichnofossils (Mangano and Buatois, 2004). It unconformably overlies the neo-Proterozoic Puncoviscana Formation and the Cañaní (519–534 Ma) and Tipayoc (550 Ma) granites (Bachmann et al., 1987; Omarini et al., 1996; Rubiolo, 2001). It is unconformably covered by the Santa Rosita Formation with trilobite fossils (*Parabolina Argentina*, *Plicatolina*,

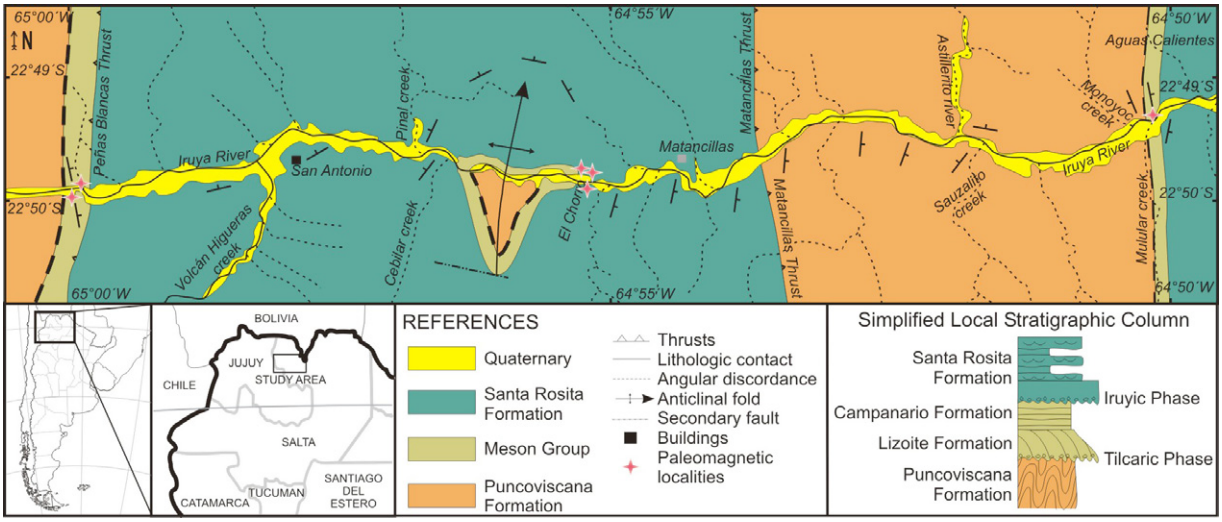


Fig. 1. Geologic map of Matancillas anticline (modified from Spagnuolo, 2004).

Parabolinella, *Angelina*) that indicate a Late Cambrian age (Marengo and Astini, 2004). The Mesón Group is subdivided into the Lizoite, Campanario, and Chalhualmayoc formations. The Lizoite Formation consists of conglomerates at the base and overlying white-pinkish quartzites and sandstones with low-angle cross-bedding stratification. The Campanario Formation is a dominantly green and purple, fine-grained, thoroughly bioturbaded succession (Mangano and Buatois, 2004) with minor fine-grained sandstones, whereas the Chahualmayoc Formation is a relatively coarse-grained, quartz-rich unit. Within the studied area, only the Lizoite and Campanario formations crop out because of the development of the Iruya unconformity that truncates the Mesón Group. Biostratigraphically dated Late Cambrian and Early Ordovician rocks of the Santa Victoria Group (Turner, 1960) overlie the Mesón Group. It is composed of sandy and shaly shallow marine packages interbedded with some conglomerates (Astini, 2003). Only the Upper Cambrian–Tremadoc (Astini, 2003) Santa Rosita Formation of the Santa Victoria Group is exposed in the study area.

3. Structural background

The studied area has been affected by three main deformational episodes: Tilcaric (Early Cambrian), Iruyic (Late Cambrian), and Andean (mainly Neogene).

The Tilcaric episode took place between 540 and 520 Ma, at the end of the Pampean orogeny, which may represent the final stages of Gondwana amalgamation (Ramos, 1999c; Sánchez Bettucci and Rapalini, 2002; Trindade et al., 2004). The Tilcaric episode generated intense deformation by the closure of the Puncoviscana Basin (Mon and Hongn, 1996; Rapela et al., 1998). The event reportedly was produced by the collision of either the Arequipa–Antofalla terrane or the Pampia block against the Río de la Plata Craton (Ramos, 1988, 1999b; Rapela et al., 1998; Rapalini, 2005). Mon and Hongn (1996) and Rubiolo (2001) identify the Tilcaric phase as responsible for notorious WNW-vergent folding with slight plunge to the north. A second phase of microfolding associated with the same phase is superimposed on the Puncoviscana Formation.

The Iruyic event occurred in the Late Cambrian as part of the Famatinian cycle (499–435 Ma) (Turner and Mendez, 1975; Rapela et al., 2001). The deformation produced by this phase was much milder than that of the Tilcaric event, generating the folding of the Mesón units during the beginning of subduction along the South American margin (Sánchez and Salfity, 1999; Astini et al., 2005). Russo (1954) and Russo and Barcat (1978) suggest the Mesón Group was already folded before the deposition of Santa Victoria Group sediments; the angular unconformity between the two units is approximately 5° (Turner and Mendez, 1975). On a regional scale, the axes of the folding have a NNE trend, and the folds have smooth geometry. This folding causes unroofing of some units of the Mesón Group in the region. Whereas at some localities, the Santa Victoria Group overlies the Chalhualmayoc Formation, in others, it overlies the Campanario or Lizoite formations, suggesting erosion of an already folded succession. The regional pinch out of the Mesón Group is tectonic and related to the Iruyic episode, and the unit might have had a much broader extent, as suggested by Aceñolaza et al. (1999).

Andean tectonics generated the uplift of the Andean Cordillera and the Pampean ranges. This cycle can be subdivided into two phases (Ramos, 1999c). During the Paleogene, the Incaic phase generated volcanism in the Puna and tectonic inversion of Cretaceous rifts. During the Neogene, the Quechua phase continued the inversion of rift structures in the Eastern Cordillera and the sub-Andean ranges. From the late Oligocene onward, retroarc continental basins were developed, and a volcanic arc developed in the Western Cordillera (Chile). Andean tectonics is responsible for the present morphostructural units in NW Argentina (Mon and Salfity, 1995): the Puna, the Eastern Cordillera, and the sub-Andean ranges.

Within the study area, Dávila (1998) identifies three main thrusts (Fig. 2). The Matancillas backthrust has a brecciated zone and places Precambrian strata over Ordovician rocks. The same author classifies the Matancillas anticline as a fault-propagated fold in response to the blind backthrust of El Cebilar. The Peñas Blancas thrust has a normal vergence to the east and places Cambrian and Precambrian over Ordovician rocks (Fig. 2).

The main structure in the study area is the Matancillas Anticline (Figs. 1 and 2), a several kilometers long, NNE–SSW structure. It is interpreted as a fault-propagated fold generated during Andean deformation. The axes of the anticline strike N26°E and plunge 10° to the north, with a steep (75°) western limb and a gently dipping (17°) eastern one (Spagnuolo, 2004).

4. Petrography

Microscopic observations of thin sections of the Mesón Group indicate that the Lizoite Formation (Fig. 3a–d) is composed of quartz-rich sandstones (modified Dott classification in Pettijohn et al., 1972) with extremely reduced poral spaces. The mean grain size is medium sand (0.3 mm); the average grain size ranges between 0.18 and 0.95 mm, indicating poor sorting. Maximum grain size of isolated grains reaches 2 mm. The sandstones contain well over 95% quartz and plot as quartz arenites. Minor percentages of rock fragments, micas, zircon, tourmaline, and sphene are present. Grains appear well rounded, though secondary quartz overgrowths prevent detailed observations. Both monocrystalline and policrystaline quartz are present. Among quartz grains, normal extinction is dominant, and both fluid and biotite inclusions, as well as opaque minerals, are present. The cement (5–15%) has a granular to microgranular texture. Occasional clay aggregates and microgranular silica is locally present.

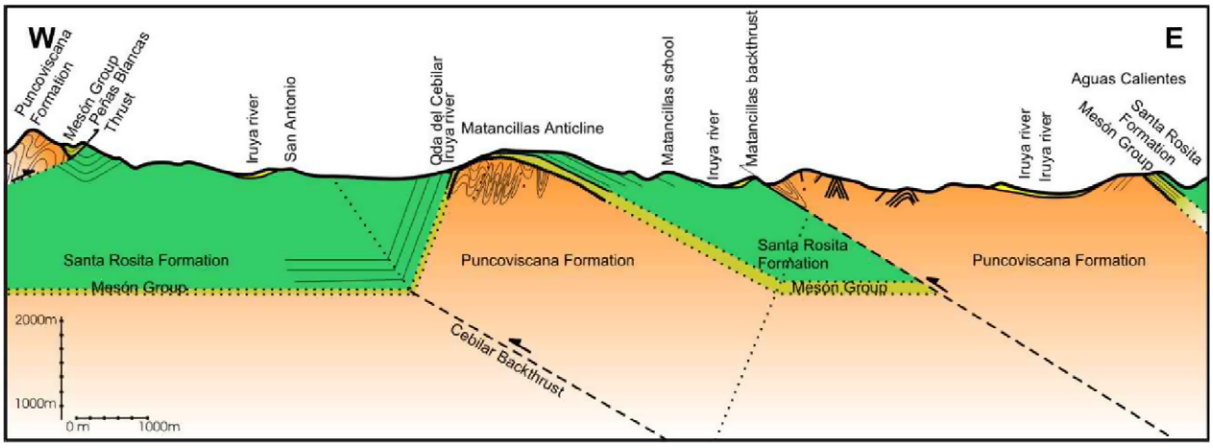


Fig. 2. Structure of the studied region. References in Fig. 1.

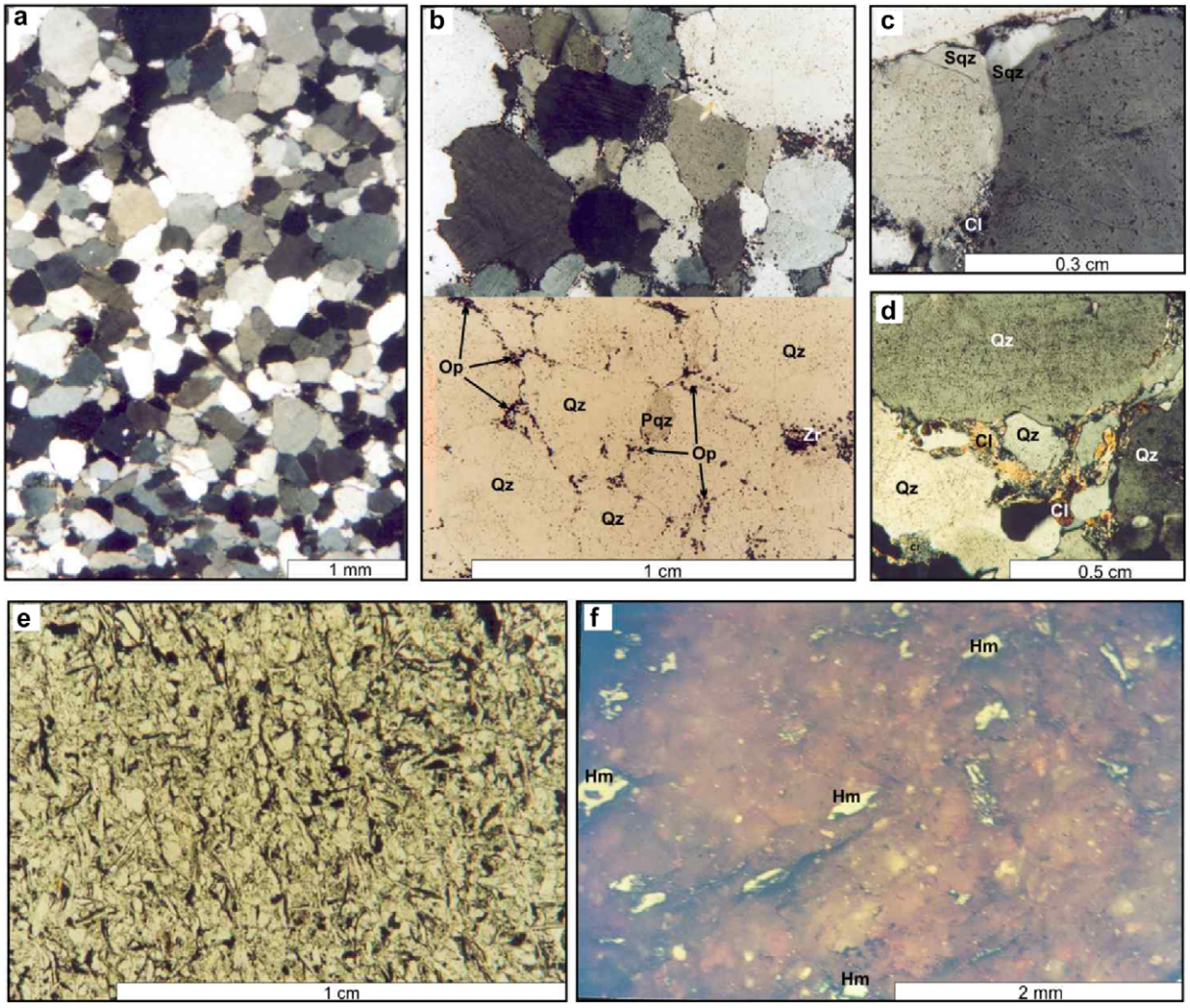


Fig. 3. Representative photomicrographs of the Mesón Group (polished sections and oil inmersion). Qz, quartz; Pqz, policrystaline quartz; Sqz, secondary quartz; Op, opaque minerals; Zr, zircon; Cl, clays; Hm, hematite. (a) Good roundness and bad selection of the Lizoite Formation. (b–d) Pore-filling opaque minerals, clay, and overgrowth quartz of the Lizoite Formation. (e and f) Campanario Formation.

The Campanario Formation (Fig. 3e and f) records much finer-grained rocks with a higher silty–shaly component. Greywacke textures with abundant mica flakes and a banded texture (grain size and amount of opaque minerals decrease from bottom to top) characterize this unit. The average size of the fragments is silt (0.03 mm). Detrital

grains are mainly quartz, muscovite, feldspar, and accessory minerals such as titanite and turmaline. Diagenesis generated dominant silica and profuse opaque mineral cementation.

Hematite, which fills the interstitial spaces, is well crystallized with grains from less than 0.01 to 0.2 mm (Fig. 3f). It presents an incipient cleavage lineament parallel to the elongation of the grains in most cases. Hematite generally occurs as fine-grained intergrowths cementing micas and quartz grains.

5. Paleomagnetic sampling and laboratory procedures

The 112 oriented samples were collected at 18 sites in the Mesón Group (Middle–Upper Cambrian) with a portable gasoline-power drill; 89 samples (14 sites) correspond to the Lizoite Formation and 23 (4 sites) to the Campanario Formation. Whenever possible, samples were oriented by means of both sun and magnetic compasses. At the laboratory, each core was sliced into one to three standard paleomagnetic specimens (2.2 cm high, 2.54 cm diameter).

5.1. Paleomagnetic study

The specimens were processed with a DC-SQUID (2G-750R) cryogenic magnetometer, and 38 pilot samples (28 from the Lizoite Formation and 10 from the Campanario Formation) were submitted to detailed stepwise thermal demagnetization (20–720 °C in 20 steps). The bulk susceptibility was measured after each step to control possible chemical changes of the samples. Heating of the samples continued until less than 10% of the initial magnetization remained. The unblocking temperatures (680 °C) show that hematite is the principal magnetic carrier, and a mineral component with unblocking temperatures less than 350 °C (phyrrotite?) was present in few samples. Eleven samples of the Lizoite Formation and two of the Campanario Formation were analyzed with stepwise alternating field (0–150 mT) demagnetization. Samples from the Lizoite Formation generally show a random distribution of the remanence direction. The Campanario samples show stable magnetic directions, but initial magnetism cannot be completely eliminated. Because of these results, thermal demagnetization was the preferred technique, and 66 more samples of the Mesón Group were demagnetized in 11 steps from 20 to 720 °C.

The magnetic behavior of each specimen was analyzed by visual inspection of Zijderveld plots, stereographic vectorial projections, and intensity demagnetization curves. After visual selection, magnetic components were determined by principal component analysis (Kirschvink, 1980), with maximum angular deviation (MAD) values under 14° for the Lizoite samples (Table 1) and under 7° for the Campanario samples (Table 2A). In most cases, components were determined with more than three consecutive demagnetization steps.

5.2. Bulk susceptibility

The initial susceptibility values of the Lizoite Formation samples were between 2×10^{-5} SI and 5×10^{-5} SI, whereas the Campanario values are a little higher from 9×10^{-6} SI to 3×10^{-4} SI. These low values can be interpreted as reflecting low concentrations of para- and ferromagnetic minerals.

5.2.1. Lizoite Formation

Sites IM3, IM15 and IM17 have maximum unblocking temperatures lower than 350 °C, suggesting iron sulfurs (likely phyrrotite) as the magnetic carrier. In IM1 and IM6, no consistent direction could be isolated. In IM10 and IM18, components of low temperature were isolated. In IM2, IM4, IM5, IM7, IM8 and IM9, components of low and high temperatures could be isolated (Fig. 4). However, they are randomly distributed both in situ and after bedding correction (Table 1 and Fig. 4).

5.2.2. Campanario Formation

The samples of the Campanario Formation show good magnetic behavior. In most cases, two components were isolated: component A between 0 and 500 °C and component B between 500 and 680 °C (Fig. 5). High unblocking temperatures (680 °C) suggest hematite as the principal ferromagnetic carrier of the later.

Component A directions are moderately well grouped with intermediate positive inclinations pointing to the west (in situ). Both in situ and after bedding correction directions do not resemble any South American Phanerozoic direction. This fact and its relatively low to moderate blocking temperatures suggest that component A could be a mixture of component B and a secondary component, not properly isolated.

Component B could be isolated in all but one sample (IM 13-1a) and shows a good directional consistency. Isolated at the highest temperatures with linear trends toward the origin in the Zijderveld plots, it is considered the characteristic remanence of the Campanario Formation. All except one sample show moderate to high positive inclinations pointing east. The remaining sample (IM 12-1a) presented a nearly antipodal direction (Table 2A).

In situ mean sample direction (Fig. 5A) is Dec = 85.8°; Inc = 76.0°; $n = 26$; $k = 7.65$; $\alpha_{95} = 11.0°$; and $R = 22.73$. After bedding correction, a slightly better grouping is observed: Dec = 81.8°; Inc = 43.6°; $n = 26$; $k = 17.28$; $\alpha_{95} = 7.0°$; and $R = 24.55$ (Fig. 5B). Opposite polarities in samples of the same site (IM 12) suggest that a sample basis mean is probably more appropriate. However, the more conservative mean site directions are also shown in Table 2A, which indicate very similar results. Improvement of the statistical parameters after bedding correction suggests a pretectonic magnetization, though statistical fold-tests (McElhinny, 1964; McFadden, 1990) are not conclusive.

Table 1
Characteristic remanence magnetization for samples of the Lizoite Formation; a is the component of low temperature, and b is the component of high temperature. Sites 2, 4, 5, 7, 8 weere also corrected by the fold axis plunge (26°N 10°E)

Sample	In situ			Bedding correction		Bedding corrected		Paleomagnetic component
	Dec (°)	Inc (°)	MAD (°)	Strike (°)	Dip (°)	Dec (°)	Inc (°)	
IM2-2a	13.4	37.0	5.9	344.0	29.5	28.2	19.3	a
IM2-3a	63.8	−29.2	3.4	344.0	29.5	58.8	−58.0	b
IM4-1b	64.9	−17.6	4.7	356.0	29.0	58.7	−44.1	a
IM4-2a	30.3	−50.1	7.8	356.0	29.0	351.3	−57.8	a
IM4-3a	37.8	−67.2	6.7	356.0	29.0	319.9	−68.7	a
IM4-4a	53.8	−24.9	8.9	356.0	29.0	41.3	−47.8	a
IM4-4a	83.0	−32.1	7.5	356.0	29.0	82	−61.0	b
IM5-2a	49.2	−13.3	10.4	356.0	29.0	41.6	−35.4	b
IM5-3a	137.6	−6.8	7.2	356.0	29.0	145.5	−23.7	b
IM5-3a	89.0	−63.9	5.6	356.0	29.0	243.0	−86.8	a
IM5-4a	137.9	50.3	6.8	356.0	29.0	122.0	28.8	a
IM5-4a	52.2	−14.9	4.6	356.0	29.0	44.3	−37.9	b
IM5-6b	46.5	−11.8	7.5	356.0	29.0	39.3	−33.0	b
IM5-6b	285.5	7.7	7.8	356.0	29.0	291	34.8	a
IM5-7a	273.2	15.3	6.9	356.0	29.0	276.9	44.0	b
IM7-1a	251.0	−28.8	13.4	344.0	31.5	253.2	2.7	b
IM7-2a	313.7	−0.2	10.3	344.0	31.5	319.3	15.1	a
IM7-2a	268.5	43.4	3.1	344.0	31.5	292.9	72.4	b
IM7-3a	239.4	20.2	8.1	344.0	31.5	234.1	50.3	b
IM7-4a	156.3	6.6	1.8	344.0	31.5	155.9	1.6	a
IM7-4a	1.9	34.7	10.7	344.0	31.5	19.1	20.7	b
IM7-5a	181.5	−11.5	8.9	344.0	31.5	186.7	−0.9	a
IM7-5a	120.6	36.3	5.0	344.0	31.5	112.7	12.4	b
IM7-6a	147.7	40.4	7.1	344.0	31.5	130.4	26.2	a
IM7-6a	282.9	35.6	5.8	344.0	31.5	308.2	60.3	b
IM8-1a	36.6	−16.4	7.3	345.0	22.3	30.9	−33.1	b
IM8-1a	77.1	−28.0	8.7	345.0	22.3	79.2	−50.3	a
IM8-2a	28.1	−2.6	4.8	345.0	22.3	26.4	−17.5	b
IM8-5a	25.4	−10.1	2.9	345.0	22.3	21.2	−23.9	b
IM8-6a	359.3	−28.6	5.5	345.0	22.3	347.6	−31.7	b
IM8-7a	73.9	33.2	9.5	345.0	22.3	75.3	10.9	a
IM8-7a	34.0	−1.1	7.1	345.0	22.3	32.8	−17.7	b
IM8-4a	17.9	−26.6	11.0	345.0	22.3	6.7	−36.8	a
IM8-4a	185.3	3.9	9.4	345.0	22.3	183.7	11.2	b
IM9-2b	198.1	−56.3	9.4	344.0	56.0	226.0	−12.0	a
IM9-2b	111.4	−34.4	7.7	344.0	56.0	175.5	−59.2	b
IM9-3b	102.9	−34.1	11	344.0	56.0	172.4	−66.1	a
IM9-3b	15.3	−14.5	9.6	344.0	56.0	349.1	−33.8	b
IM9-4a	205.3	−13.3	5.6	344.0	56.0	200.9	23.8	a
IM9-4a	40.5	−64.1	4.2	344.0	56.0	278.0	−53.6	b
IM9-5a	319.9	10.6	13.7	344.0	56.0	339.4	25.8	a
IM9-5a	356.8	14.9	7.8	344.0	56.0	3.5	−1.9	b
IM9-7a	343.9	−2.5	2.9	344.0	56.0	341.9	−1.3	a
IM9-7a	24.7	−1.4	10.6	344.0	56.0	8.4	−33.6	b
IM10-1a	176.3	−1.1	5.9	340.0	55.0	170.4	12.6	a
IM10-3a	246.4	19.2	1.6	340.0	55.0	237.7	73.9	a
IM10-4a	190.4	4.3	6.9	340.0	55.0	174.8	27.2	a
IM18-7a	174.4	−24.7	7.4	345.0	45.7	189.2	−10.7	a
IM18-4a	12.6	−36	8.8	345.0	45.7	332.5	−42.7	a
IM18-1a	148.4	1.5	7.5	345.0	45.7	152.2	−10.7	a
Mean Comp a	D = 112.4° $n = 26$	I = −45.4° $R = 5.19$	$\alpha_{95} = 60.7°$ $k = 1.2$			D = 137° $n = 26$	I = −54.0° $R = 3.73$	$\alpha_{95} = 76.2°$ $k = 1.12$
Mean Comp b	D = 31.7° $n = 24$	I = 10.7° $R = 9.1$	$\alpha_{95} = 39.4°$ $k = 1.54$			D = 18.5° $n = 24$	I = −32.0° $R = 8.77$	$\alpha_{95} = 40.7°$ $k = 1.51$

Further confirmation of a pretectonic magnetization exists because if it were posttectonic, the remanence should be of Late Tertiary or younger in age, but the in situ declination and inclination values are anomalous with respect

Table 2A
Characteristic remanence magnetization for samples of the Campanario Formation

Sample	In situ			Bedding correction		Bedding corrected remanence		
	Dec (°)	Inc (°)	MAD (°)	Strike (°)	Dip (°)	Dec (°)	Inc (°)	MAD (°)
IM11-1a	72.8	76.8	1.7	356	37	82.1	40.1	1.7
IM11-2a	134.8	80.7	2.9	356	37	96.2	46.4	2.9
IM11-3a	17.9	82.2	3.8	356	37	74.8	49.5	3.8
IM11-3b	5.8	82.1	4	356	37	73.6	51	4
IM11-4a	218.8	74.6	5.4	356	37	110.1	61.5	5.4
IM11-5b	144.3	82.2	2.8	356	37	96.0	48.4	2.8
IM11-6a	141.3	57.4	4.7	356	37	116.5	29.2	4.7
IM11-7a	106.2	63.4	6.1	356	37	96.0	27.5	6.1
IM11-8a	131.5	72.9	4.7	356	37	101.8	39.7	4.7
IM12-1a	133.5	−64.2	4.4	356	37	219.7	−63.6	4.4
IM12-2a	72.7	79.9	6.2	356	37	82.8	43.1	6.2
IM12-2b	44.3	81.1	3.5	356	37	77.5	46.0	3.5
IM12-3a	129.6	87.7	3.6	356	37	88.5	51.3	3.6
IM12-3b	237.1	76.3	5.8	356	37	101.3	64.2	5.8
IM12-4a	109.5	82.1	4	356	37	90.5	45.7	4
IM12-5a	286.8	85.3	3.3	356	37	82.9	57.4	3.3
IM12-6a	244.5	71.6	3.8	356	37	105.0	69.2	3.8
IM12-7a	293.5	75.1	4.8	356	37	69.4	65.4	4.8
IM13-2a	85.1	60	6.6	344	18.5	81.4	41.7	6.6
IM14-1a	56.7	62.9	2.7	341	31	63.3	32.4	2.7
IM14-1b	65.9	62.1	4.7	341	31	68.2	31.2	4.7
IM14-2a	62.6	55	2.1	341	31	65.7	24.2	2.1
IM14-3a	65.2	51.2	4.1	341	31	67.1	20.3	4.1
IM14-4a	70.6	49.6	3.6	341	31	70.7	18.6	3.6
IM14-4b	72.2	49.3	6.6	341	31	71.8	18.3	6.6
IM14-5a	75.5	50	4.5	341	31	74.1	19.1	4.5
Mean	D = 85.8°	I = 76.0°	α_{95} = 11.0°			D = 81.8°	I = 43.6°	α_{95}=7.0°
	n = 26	R = 22.73	k = 7.65			n = 26	R = 24.55	k = 17.28

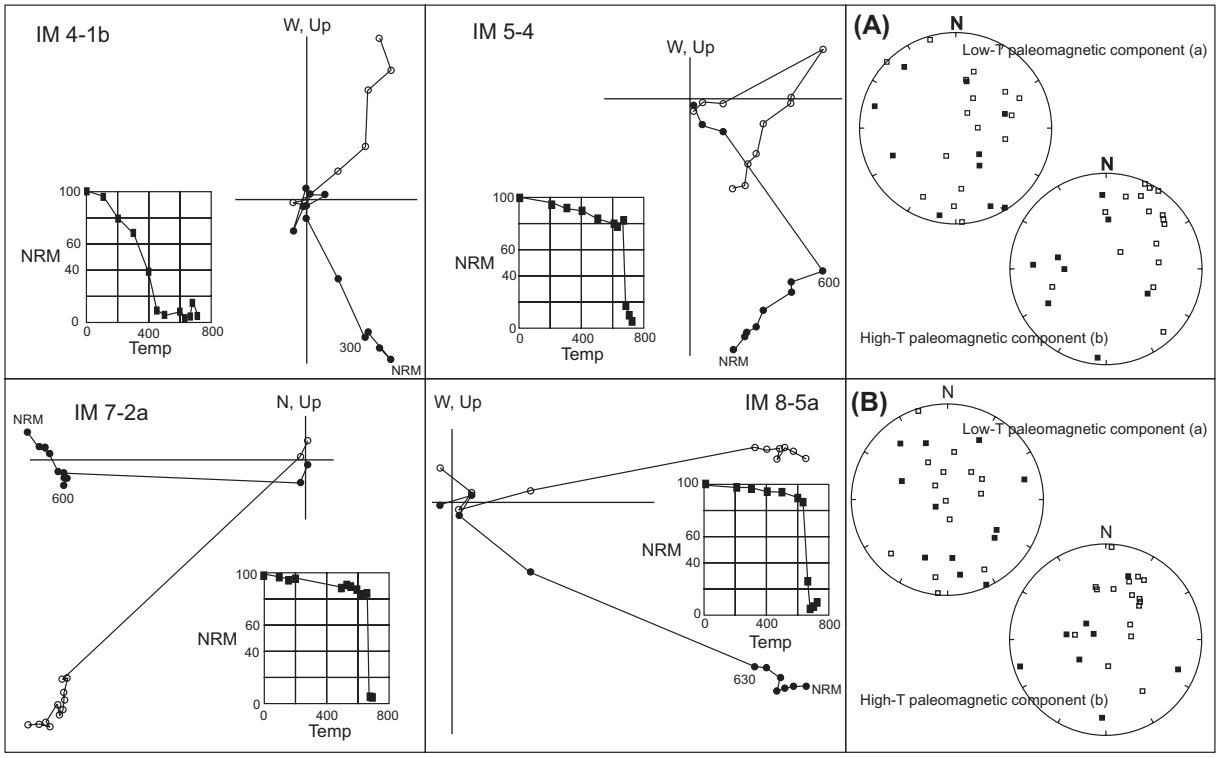


Fig. 4. Examples of magnetic behavior for four pilot specimens of the Lizoite Formation; open (closed) symbols represent vertical (horizontal) projection. Equal-angle stereoplot, open (closed) symbol is negative (positive) inclination. (A) In situ distribution of sample directions. (B) After bedding correction distribution of sample directions.

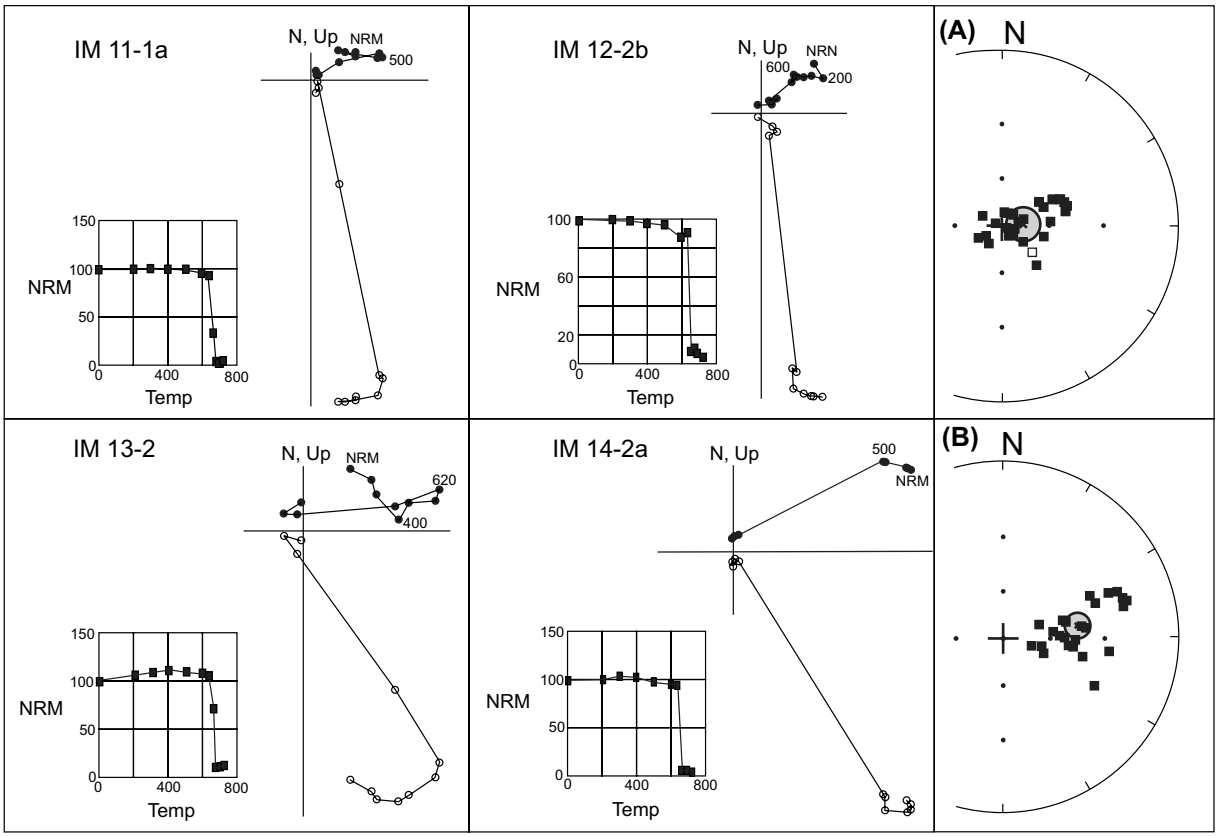


Fig. 5. Typical demagnetization behavior of the Campanario Formation samples represented by Zjiderveld plots and demagnetization curves. In the vector diagram, closed (open) symbols represent projections on the horizontal (vertical) plane. Equal-angle stereoplot, open (closed) symbol is negative (positive) inclination. (A) In situ distribution of sample directions. (B) After bedding correction distribution of sample directions.

Table 2B
Site mean remanence directions for the Campanario Formation; *N*, number of samples

Site	*n*	In situ			Bedding correction		Bedding corrected remanence		
		Dec (°)	Inc (°)	α_{95} (°)	Strike (°)	Dip (°)	Dec (°)	Inc (°)	α_{95} (°)
IM11	9	123	79.9	9.7	356	37	94.5	44.6	9.7
IM12	9	290.4	85.3	8.7	356	37	82.4	57.2	8.7
IM13	1	85.1	60	–	344	18.5	81.4	41.7	–
IM14	7	67.6	54.4	5.2	341	31	68.8	23.4	5.2
Mean	N = 4	D = 79.9°	D = 72.9° *R* = 3.83	α_{95} = 22.6° *k* = 17.54			D = 80.8°	I = 42.2° *R* = 3.88	α_{95} = 18.6° *k* = 25.36

to the expected Late Tertiary direction at the study locality. Magnetic fabric and petrographic studies also point to a primary magnetization.

A paleomagnetic pole was computed on the basis of the bedding corrected characteristic remanence direction, FC: 4.5°S, 359.0°, dp = 5.5, dm = 8.8, *n* (samples) = 26. Tables 2A and 2B show the magnetic parameters of samples of the Campanario Formation.

5.3. Anisotropy of magnetic susceptibility (AMS)

Anisotropy of magnetic susceptibility (AMS) was measured for 8 samples of the Campanario Formation with a Minisep at the laboratory of IAG (Universidade de São Paulo, Brasil) to determine if there was any preferential alignment of the magnetic minerals that could influence the recording of the Earth's magnetic field. Those rocks have very low susceptibilities because of the abundance of diamagnetic minerals, so a correction for the diamagnetic contribution was made to the data, as shown in Hongn et al. (1999).

After correction, the minimum principal axis (K3) is nearly vertical, whereas K1 and K2 are distributed on the bedding plane. This result strongly suggests a pretectonic, likely depositional or diagenetic, fabric and supports the case for a primary origin of magnetization.

5.4. Isothermal remanent magnetization (IRM)

To better characterize the magnetic carriers, isothermal remanence stepwise acquisition experiments were performed. These experiments were carried out with an ASC (Model

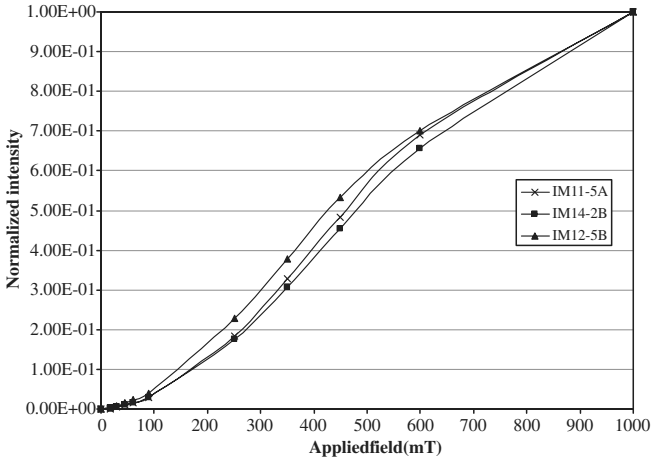


Fig. 6. IRM acquisition curves in the pelites from the Campanario Formation. Normalized intensities versus field strength.

Table 3
Comparison of applied field and isothermal remanence mean inclination values with respect to the bedding plane

Sample	Applied Field Inc (°) (Tectonic coordinates)	IRM Inc (°) (Tectonic coordinates)	Standard deviation
IM 11-5a	44.0	50.0	5.4°
IM12-5b	44.8	41.1	11.1°
IM14-2	44.5	49.9	1.1°

No significant shallowing of inclination. IRM inclination is an average of the values obtained with an applied field of 1000, 1017, 1030, and 1043 mT.

IH10-30) pulse magnetizer. The curves of remanence acquisition show that saturation is not reached at 1 T, indicating that hematite is the principal remanence carrier (Fig. 6). The very flat slope up to 100 mT (Fig. 6) indicates a lack of any significant ferrimagnetic contribution. Because AMS parameters provide no insight into any possible distortion of the magnetic field recorded by the Campanario samples, due to compaction-induced flattening of the remanence (Tan and Kodama, 2002), a test was performed applying IRM at fields around 1 T along a direction nearly 45° with respect to the bedding plane. Any significant systematic deflection of the IRM toward the bedding plane would suggest that a correction of the characteristic remanence directions is needed (Hodych and Buchan, 1994). This procedure was applied to three samples of the Campanario Formation.

Table 3 shows the measured values of inclination of IRM with respect to the bedding plane. There is no systematic inclination shallowing; therefore, the characteristic mean remanence direction of the Campanario Formation can be considered an accurate record of the mean paleomagnetic field.

6. Interpretation of results

The rocks of the Mesón Group, particularly those of the Lizoite Formation, have a very mature mineralogy (95% quartz) but very poor sorting (Fig. 3a). Mineralogical maturity and relatively well-rounded grains of the sandstones can be attributed to either protracted recycling or first-cycle products under particular climatic conditions (Dott, 2003). However, the large range in grain sizes (poor sorting) and scarcity of inherited older growth rims within quartz grains, together with the great contrast with the underlying folded units and the relatively short gap that separates the Mesón Group from the underlying Puncoviscana Formation and late-stage intrusive, suggest little recycling and mechanical rework. Sandstones with extremely mature mineralogy presently formed in low latitudes through intense chemical alteration and first-cycle shallow marine sandstones (Johnsson et al., 1988, 1991; Dott, 2003). Low latitude wet and warm climates are particularly prone to generate first-cycle, quartz-rich arenites as a product of chemical weathering (Dott, 2003). Such warm/subtropical conditions of deposition are supported by the paleomagnetic pole, which indicates that the Mesón Group sediments were deposited at 25.7° (−5.1°/+5.9°)S. The minimal porosity within the rock indicates deep burial and compaction but no metamorphism, as suggested by the recrystalization patterns on quartz grains and their normal extinction. Considering the bracketed stratigraphic interval of the Campanario Formation and accepting a primary magnetization, an approximate age of 500 Ma for the Campanario paleomagnetic pole is inferred.

Using the reconstruction parameters of Lottes and Rowley (1990), the Campanario Formation paleomagnetic pole was rotated into African coordinates: 24.0°N, 36.9°E. This paleomagnetic pole is the first obtained from Cambrian rocks of Argentina, widely accepted as authoctonous and part of Gondwana in the Cambrian. The position in northeastern Africa is roughly consistent with some other Cambro–Ordovician poles of Gondwana (Fig. 7; see Meert and Van der Voo, 1996; D'Agrella-Filho et al., 2000; Meert et al., 2003; Trindade et al., 2004). However, it is displaced with respect to coeval average poles of Gondwana (see Grunow, 1999; Rapalini and Cingolani, 2004) (Fig. 8). At least three alternative explanations can be explored for this pole position:

6.1. Alternative 1

Taking the Campanario Formation pole as wholly representative of Gondwana, the APWP of this supercontinent should be more complex than those proposed in the last decade (Fig. 8; Meert, 2001). Numerous different Early Paleozoic APWPs have been postulated for Gondwana in the last 30 years. Most authors tend to agree with a rather simple path that starts in the Late Ediacaran (~550 Ma) over the SW margin of Gondwana and ends by the Early Ordovician in northern Africa (e.g., Meert and Van der Voo, 1996; Grunow, 1999; Meert, 2001; Trindade et al., 2006). If the Campanario Formation pole represents the Gondwana pole around 500 Ma, it would

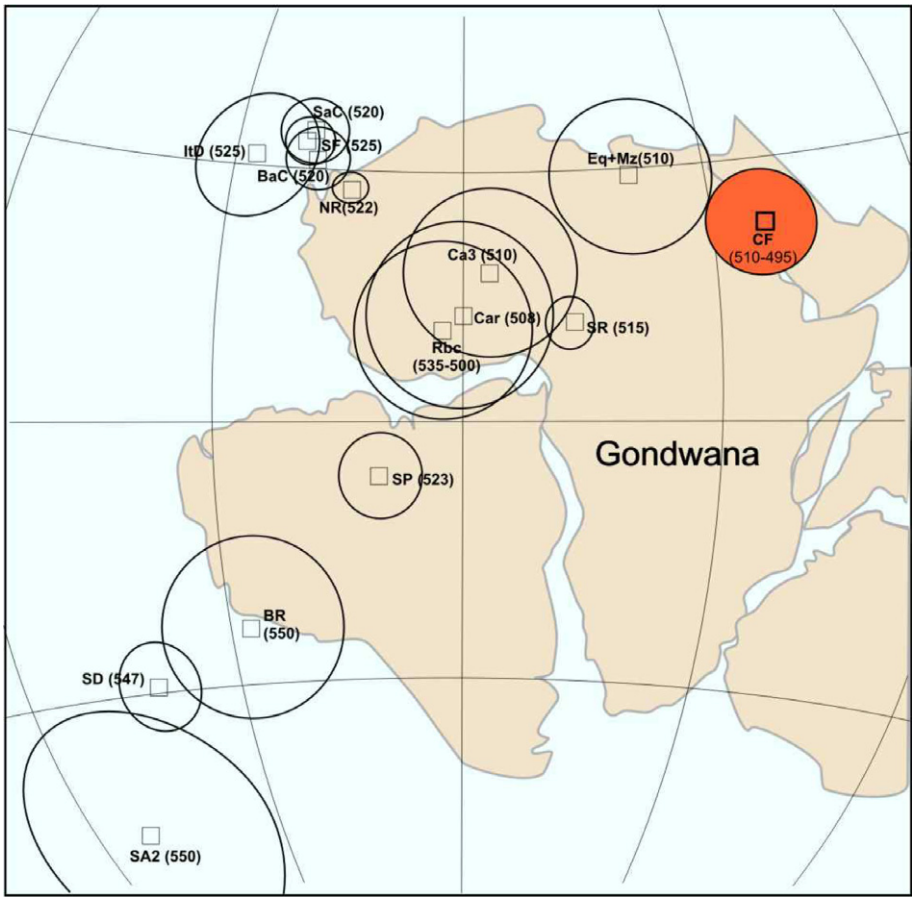


Fig. 7. Poles from Gondwana in African coordinates. Ages in Ma (D'Agrella-Filho et al., 2000, 2004; Meert, 2001; Meert et al., 2003; Sánchez Bettucci and Rapalini, 2002; Trindade et al., 2004, 2006; Gose et al., 2004), CF, this work; Eq + Mz, Equeefa and Mzumbe; SD, Sinyai Dolerite; BR, Bhander and Rewa; CA3, central Australian mean pole; SP, Salt Pseudomorphs; NR, Ntonya ring structures; SF, Salitre Formation; SR, Sor Rondane intrusions; RBc, Juiz de Fora Complex; Sac, Salitre Formation; BaC, Bambuí carbonates; Car, Carion granite pluton; ItD, Itabaiana dikes; SA2, Sierra de las Ánimas magmatic complex.

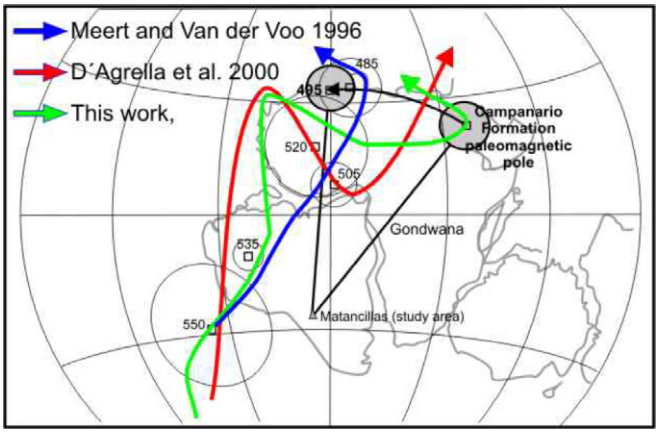


Fig. 8. Average poles from Gondwana (Lower Paleozoic) (Grunow, 1999; Meert, 2001; Rapalini and Cingolani, 2004). Campanario paleomagnetic pole and its relationship with the average pole from Gondwana of approximately the same age. Apparent polar wander paths: Meert and Van der Voo, 1996, grey arrow; D'Agrella-Filho et al., 2000; black arrow and this work, dotted arrow.

imply an extra loop on the path from central northern Africa around 510 Ma to the Red Sea around 500 Ma, turning back to northern Africa by approximately 490 Ma. Such large displacements would imply high drift velocities of Gondwana of almost 6000 km in 20 m.y. between 510–490 Ma. The maximum speed limit of continental plates is highly debatable. Different authors suggest variable plate speeds from 11 to 38 cm/y (Gordon, 1991; Gurnis and Torsvik, 1994; Evans, 1998; Sánchez Bettucci and Rapalini, 2002; Trindade et al., 2004, 2006). The inferred velocity for this hypothetical movement is close to the upper limit of any previously proposed drift for Gondwana (around 30 cm/a). The Campanario Formation pole falls near to two other Cambrian poles, the Piquete Complex in the São Francisco Craton in Brazil, assigned to 510 Ma (D'Agrella-Filho et al., 1986), and a recent pole for the Kalahari Craton (Gose et al., 2004) proposed to be approximately 530 Ma. In this case, the other two poles should be slightly younger (i.e., around 500 Ma) than originally proposed by the respective authors. Because this interpretation is based on only three paleomagnetic poles of probable different ages and implies extremely fast drift rates, this explanation should be considered unlikely.

6.2. Alternative 2

The second alternative implies a relative movement between the study locality and cratonic areas of South America since the acquisition of magnetization. The Mesón Group is deposited on the so-called Pampia block (see Ramos, 1988; Rapalini, 2005). This block, with its possible authocthonous (Rapela et al., 2001), allochthonous (Rapela et al., 1998), or Amazonian affinities (Rapalini, 2005), apparently collided with the Río de la Plata Craton (already part of the recently formed Gondwana) in the Early Cambrian (~530 Ma). Therefore, no significant displacements should be expected from the approximately 500 Ma pole of the Campanario Formation. However, Simpson et al. (2003) and Schwartz and Gromet (2004) speculate possible northward displacements along a "transform-like" margin between Pampia and the rest of Gondwana in the Early Ordovician. Speculatively, this displacement could be related to the final steps of amalgamation of the Kalahari Craton, which is best expressed by first-order N–S sinistral transcurrent faults in eastern Uruguay (Basei et al., 2000; Campos Neto, 2000).

The available poles of Gondwana do not allow evaluations of a few hundred kilometers of displacement of Pampia relative to Gondwana in the Early Ordovician. In any case, the recently obtained Cambrian paleomagnetic pole for the Kalahari Craton (Gose et al., 2004) is misplaced in the Gondwana path if its proposed 530 Ma age is valid. In this case, a northward displacement and clockwise rotation of Kalahari since the Early Cambrian should follow. However, a younger age around 500 Ma for this Kalahari pole is more consistent with the Gondwana database. The possibility of significant displacements of some of the southwestern Gondwana blocks during the final stages of supercontinent assembly should be investigated. Moreover, the apparent systematic clockwise rotations found on Early Ordovician volcanics in the western margin of Pampia (Conti et al., 1996; Spagnuolo et al., in press) argue in favor of regional readjustments that should be addressed.

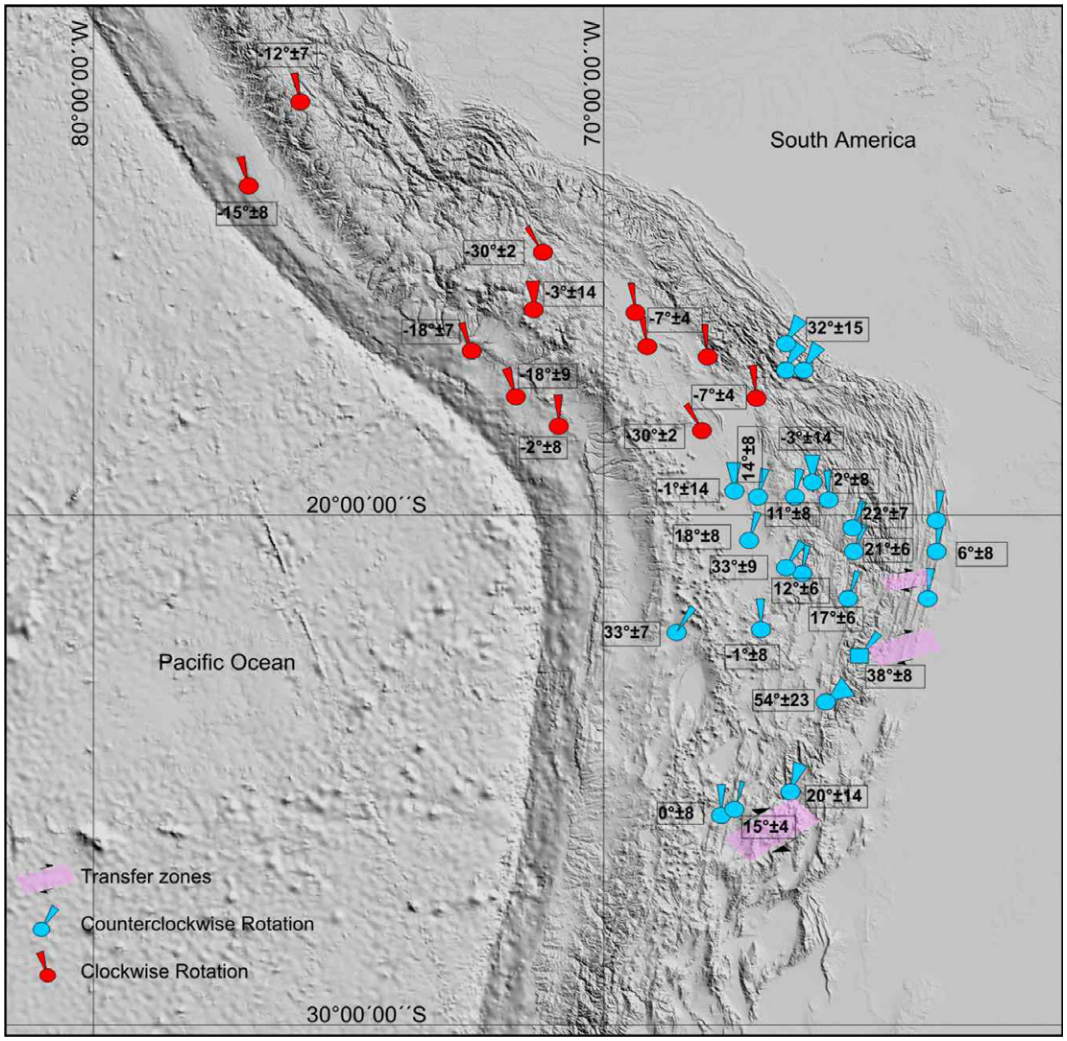


Fig. 9. Satellite image from the Central Andes showing clockwise (black dots; square: study area) and counterclockwise (white dots) paleomagnetically determined rotations around vertical axes over Mesozoic and Tertiary units. Numbers indicate the magnitude of rotation (for more details, see Lamb, 2001; Richards and Butler, 2004; Somoza and Tomlinson, 2002). The grey blocks represent transfer zones (Kley, 1996; Kley and Monaldi, 1999).

6.3. Alternative 3

Taking the 495 Ma mean pole of Gondwana as a reference (34°N, 0°E, $\alpha_{95} = 6°$; Grunow, 1999), a significant clockwise rotation of 37.8° ± 7.6° (Fig. 8) with an insignificant inclination anomaly of 4.2° ± 6.8° is indicated by the Campanario Formation pole (Beck et al., 1986). The simplest explanation for this rotation is its relation to the Central Andes rotation pattern (CARP, Somoza et al., 1996), which affects many Mesozoic and Cenozoic units along the South American margin (Fig. 9). The observed sense and magnitude are consistent with rotations recorded in Mesozoic and/or Cenozoic rocks in the region.

Previous paleomagnetic studies indicate that rotations of this magnitude and sense associated with the Bolivian Orocline (CARP) may have affected the present studied area (Somoza and Tomlinson, 2002). Therefore, the Campanario Formation pole may reflect a regional or local clockwise rotation associated to the development of the Bolivian Orocline (see Fig. 9) (Spagnuolo et al., 2005). Richards and Butler (2004) performed a paleomagnetic analysis of Mesozoic and Cenozoic sedimentary strata of Bolivia and found clockwise rotations of approximately 25° for areas south of the axis of the Bolivian Orocline at a similar longitude and some 200 km north of our area.

However, Kley and Monaldi (1999) describe the presence of transfer zones in the Andean margin in northern Argentina and southern Bolivia, associated with the eastward migration of the Andean orogenic front. According to Kley (1996), the existence of structural "obstacles" associated with a Cretaceous rift that restrict the eastward propagation of anticlines in the sub-Andean region served as transfer zones. In this interpretation, the Matancillas anticline is located along a dextral transfer zone that produced clockwise deflections of the structural trends, as seen in the strike of the main fold axes. Thus, a local block rotation associated with these structures may have caused the observed paleomagnetic rotation. A combination of a rotation associated to a regional rotation pattern (orocline) plus a local component due to dextral displacements along transfer zones is also possible.

Paleomagnetic studies of Mesozoic and Cenozoic rocks in the study area, as well as further studies on other outcrops of the Mesón Group, are needed to rule out or confirm any of the three alternatives. In the meantime, the more conservative third alternative is considered the most likely.

7. Conclusions

A reconnaissance paleomagnetic study was carried out on the Middle–Late Cambrian Lizoite and Campanario formations exposed along the Iruya River near the Matancillas anticline on the eastern boundary of the Eastern Cordillera, province of Salta, Argentina. Paleomagnetic results of the Lizoite Formation were scattered, but the Campanario Formation reveals a stable and primary magnetic remanence, which permits the calculation of a paleomagnetic pole (4.0°S, 359.0°E). A paleolatitude of approximately 26° is inferred for deposition of the Campanario Formation sediments, which is in agreement with evidence of enhanced chemical weathering in warm and wet climates, as inferred from petrographical arguments and considerations. The paleomagnetic pole is not consistent with the Cambrian–Ordovician poles of Gondwana, indicating an anomaly of 38° ± 8° in the declination value. This finding may be explained by (1) a more complex Gondwana APWP, (2) relative movements of the Pampia block with respect to Gondwana, or (3) a clockwise rotation of the study area of around 30°, related to the regional deflections associated to the Bolivian Orocline and/or to local crustal block rotations related to a dextral transfer zone along the Andean front. Considering the available paleomagnetic and geologic evidence, the last explanation is considered the most likely.

Acknowledgments

This study was supported through grants by the Universidad de Buenos Aires, UBACyT X262, and Conicet PIP-5783 to Augusto Rapalini. SuperIAPD and GMAP software packages by Torsvik (NGU) were used to analyze the data, as well as the Estereográfica-GR 1.0 by E. Cristallini (UBA). Luis Uberti and Larisa Marengo collaborated during the fieldwork. Comments by reviewers M. D'Agrella-Filho, Z.X. Li, and editor V. Ramos significantly improved the final version.

References

Aceñolaza, F.G., Buatois, L.A., Mángano, M.G., Esteban, S., Tortello, M.F., Aceñolaza G.F., 1999. Cámbrico y Ordovícico del Noroeste argentino. In: Caminos, R. (Ed.), Geología Argentina, Instituto de Geología y Recursos Minerales, Anales 29, pp. 167–187.

Allmendinger, R.W., Jordan, T., Kay, S., Isacks, B.L., 1997. The evolution of the Altiplano-Puna Plateau of the Central Andes. Annual Reviews of Earth and Planetary Sciences 25, 139–174.

Astini, R.A., 2003. The Ordovician Proto-Andean Basins. In: Benedetto, J.L. (Ed.), Ordovician Fossils of Argentina. Secretaría de Ciencia y Tecnología, Universidad Nacional de Córdoba, Argentina, pp. 1–74.

Astini, R., Spagnuolo, C., Dávila, F., Rapalini, A., 2005. El Grupo Mesón y la discordancia irúyica en el noroeste argentino: un aporte para la discusión de su origen. 16° Congreso Geológico Argentino, La Plata, Argentina. Actas CD.

Bachmann, G., Grauert, B., Kramm, U., Lork, A., Miller, H., 1987. El magmatismo del Cámbrico Medio-Cámbrico Superior en el basamento del noroeste argentino: investigaciones isotópicas y geocronológicas sobre los granitoides de los complejos intrusivos Santa Rosa de Tastil y Cañaní. Actas 10° Congreso Geológico Argentino, San Miguel de Tucumán, Tomo 4, pp. 125–127.

Basei, M.A.S., Siga Jr., O., Masquelin, H., Harara, O.M., Reis Neto, J.M., Preciozzi, P.F., 2000. The Dom Feliciano Belt of Brazil and Uruguay and its foreland Domain, the Río de la Plata Craton. In: Cordani, U.G., Milani, E.J., Thomaz-Filho, A., Campos, D.A. (Eds.), Tectonic Evolution of South America, Río de Janeiro, pp. 311–334.

Beck, M., Burmester, R.F., Craig, D.E., Gromme, C.S., Wells, R.E., 1986. Paleomagnetism of Middle Tertiary volcanic rocks from the western cascade series, Northern California. Journal of Geophysical Research 91 (B8), 8212–8230.

Beck, M., 1998. On the mechanism of crustal block rotations in the central Andes. Tectonophysics 299, 75–92.

Buatois, L.A., Mangano, M.G., 2003. La icnofauna de la Formación Puncoviscana en el noroeste argentino: la colonización de fondos oceánicos y reconstrucción de paleoambientes y paleoecosistemas de la transición precámbrica-cámbrica. Ameghiniana 40 (1), 103–117.

Campos Neto, M., 2000. Orogenic Systems from Southwestern Gondwana. In: Cordani, U.G., Milani, E.J., Thomaz-Filho, A., Campos, D.A. (Eds.), Tectonic Evolution of South America, Río de Janeiro, pp. 19–40.

Conti, C.M., Rapalini, A.E., Coira, B., Koukharsky, M., 1996. Paleomagnetic evidence of an early Paleozoic rotated terrane in northwest Argentina: a clue for Gondwana–Laurentia interaction? Geology 24 (10), 953–956.

D'Agrella-Filho, M.S., Raposo, M.I.B., Egydio-Silva, M., 2004. Paleomagntic study of the Juiz de Fora Complex, SE Brazil: implications for Gondwana. Gondwana Research 7 (1), 103–113.

D'Agrella-Filho, M.S., Babinsky, M., Trindade, R.I.F., Van Schmus, W.R., Ernesto, M., 2000. Simultaneous remagnetization and U–Pb isotope resetting in Neoproterozoic carbonates of the São Francisco craton, Brazil. Precambrian Research 99, 179–196.

D'Agrella-Filho, M.S., Pacca, I.G., Sato, K., 1986. Paleomagnetism of metamorphic rocks from the Piquete region – Ribeira Valley, Southeastern Brazil. Revista Brasileira de Geofísica 4, 79–84.

Dávila, F. 1998. Proyecto Iruya, Informe Geológico. Unpublished report for IAMGOLD, 32 p.

Do Campo, M., Guevara Ribeiro, S., 2005. Provenance analysis and tectonic setting of late Neoproterozoic metasedimentary successions in NW Argentina. Journal of South American Earth Sciences 19, 143–153.

Dott, R.H., 2003. The importance of eaolian abrasion in supermature quartz sandstones and the Paradox of weathering on vegetation-free landscapes. Journal of Geology 111, 387–405.

Evans, D.A., 1998. True polar wander, a supercontinental legacy. Earth and Planetary Science Letters 157, 329–341.

Gansser, A., 1973. Facts and theories on the Andes. Journal of the Geological Society London 129, 93–131.

Gordon, R.G., 1991. Plate tectonic speed limits. Nature 349, 16–17.

Gose, W.A., Johnston, S.T., Thomas, R.J., 2004. Age of magnetization of Mesoproterozoic rocks from the Natal sector of the Namaqua-Natal belt, South Africa. Journal of African Earth Sciences 40, 137–145.

Grunow, A.M., 1999. Gondwana events and paleogeography: a paleomagnetic review. Journal of African Earth Sciences 28 (1), 53–69.

Gurnis, M., Torsvik, T.H., 1994. Rapid drift of large continents during the late Precambrian and Paleozoic: paleomagnetic constraints and dynamic models. Geology 22 (11), 1.023–1.026.

Hernández, R.M., Reynolds, J., Disalvo, A., 1996. Análisis tectosedimentario y ubicación geocronológica del Grupo Orán en el río Iruya. Boletín de Informaciones Petroleras, Tercera Serie 12 (45), 80–93.

Hodych, J.P., Buchan, K.L., 1994. Early Silurian paleolatitude of the Springdale Group redbeds of central Newfoundland: a paleomagnetic determination with a remanence anisotropy test for inclination error. Geophysical Journal International 117, 640–652.

Hongn, F.D., Arangure, A., Tubía, J.M., Mon, R., 1999. Estructura, fábrica magnética y emplazamiento de los granitos de Brealito y La Paya, basamento del valle Calchaquí, Salta, Argentina. Acta Geológica Hispánica 34 (2–3), 301–317.

Johnsson, M.J., Stallard, R.F., Meade, R.H., 1988. First-cycle quartz arenites in the Orinoco Basin, Venezuela and Colombia. Journal of Geology 96, 263–277.

Johnsson, M.J., Stallard, R.F., Lundberg, N., 1991. Controls on the comosition of fluvial sands from a tropical weathering environment: Sands of the Orinoco River drainage basin, Venezuela and Colombia. Geological Society of American Bulletin 103, 1622–1647.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophysical Journal Royal Astronomical Society 67, 699–718.

Kley, J., 1996. Transition from basement-involved to thin-skinned thrusting in the Cordillera Oriental of southern Bolivia. Tectonics 15 (4), 763–775.

Kley, J., Monaldi, C.R., 1999. Estructura de las Sierras Subandinas y del Sistema de Santa Bárbara. In: Gónzalez Bonorino, G., Omarini, R., Viramonte, J. (Eds.), XIV Congreso Geológico Argentino, Salta, Tomo I, pp. 415–425.

Lamb, S., 2001. Verticle axis rotation in the Bolivian orocline, South America, 1, Paleomagnetic analysis of Cretaceous and Cenozoic rocks. Journal of Geophysical Research 106 (10), 26605–26632.

Lottes, A.L., Rowley, D.B., 1990. Reconstruction of the Laurasian and Gondwanan segments of Permian Pangaea. Paleogeography and Biogeography, Geological Society Memoir No. 12, pp. 383–395.

Mangano, M.G., Buatois, L.A., 2004. Reconstructing Early Phanerozoic intertidal ecosystems: ichnology of the Campanario Formation in NW Argentina. Fossils and Strata 51, 17–38.

Marengo, L., Astini, R., 2004. Las concentraciones fosilíferas de la Formación Santa Rosita (Cámbrico Superior-Tremadociano), Río Iruya, Provincia de Salta, Argentina. Actas X Reunión Argentina de Sedimentología, San Luis, Argentina, pp. 95–97.

McElhinny, M.W., 1964. Statistical significance of the fold test in Paleomagnetism. Geophysical Journal Royal Astrony Society 8, 33–40.

McFadden, P.L., 1990. The fold test as an analytical tool. Geophysical Journal International 135 (2), 329–338. doi:10.1046/j.1365-246X.1998.00640.x.

Meert, J.G., 2001. Growing Gondwana and rethinking Rodinia: a paleomagnetic perspective. Gondwana Research 4 (3), 279–288.

Meert, J.G., Van der Voo, R., 1996. Paleomagnetic and $^{40}Ar/^{39}Ar$ study of the Sinyai Dolerite, Kenya: implications for Gondwana Assembly. Journal of Geology 104, 131–142.

Meert, J.G., Nédélec, A., Hall, C., 2003. The stratoid granites of central Madagascar: paleomagnetism on further age constraints on neoproterozoic deformation. Precambrian Research 120 (1–2), 101–129.

Mon, R., 1999. Cordillera Oriental. In: Gónzalez Bonorino, G., Omarini, R., Viramonte, J. (Eds.), XIV Congreso Geológico Argentino, Salta, Tomo I, pp. 426–431.

Mon, R., Hongn, F.D., 1996. Estructura del basamento proterozoico y paleozoico inferior del norte argentino. Revista de la Asociación Geológica Argentina 51 (1), 3–14.

Mon, R., Salfity, J.A., 1995. Tectonic evolution of the Andes in Northern Argentina. In: Tankard, A.J., Suárez, R.S., Welsink, H.J., Petroleum basins of South America, AAPG Memoir 62, pp. 269–283.

Omarini, R.H., Sureda, R.J., Gotze, H.J., Seilacher, A., Pfluger, F., 1999. Puncoviscana folded belt in Northwestern Argentina: testimony of Late Proterozoic Rodinia fragmentation and pre-Gondwana collision episodes. International Journal of Earth Science 88, 76–97.

Omarini, R.H., Torres, J.G., Moya, L.A., 1996. El stock Tipayoc: génesis magmática en un arco de islas del Cámbrico Inferior en el Noroeste de Argentina. Actas 13° Congreso Geológico Argentino, 3° Congreso Geológico de Hidrocarburos 5, 545–560.

Pettijohn, F.J., Potter, P., Siever, R., 1972. Sand and Sandstones. Springer-Verlag, Berlin, Heidelber, New York, 18 pp.

Ramos, V.A., 1988. Late Proterozoic-Early Paleozoic of South America-a collisional history. Episodes 11, 168–174.

Ramos, V.A., 1999a. Plate tectonic setting of the Andean Cordillera. Episodes 22 (3), 183–190.

Ramos, V.A., 1999b. Las Provincias geológicas del territorio argentino. Instituto de Geología y Recursos Minerales, Geología Argentina, Anales 29 (24), pp. 41–96.

Ramos, V.A., 1999c. Rasgos estructurales del territorio argentino. Instituto de Geología y Recursos Minerales, Geología Argentina, Anales 29 (24), pp. 715–784.

Ramos, V.A., 2000. The Southern Central Andes. In: Cordani, U.G., Milani, E.J., Thomaz Filho, A., Campos, D.A. (Eds.), Tectonic Evolution of South America. 31° International Geological Congress, Río de Janeiro, pp. 561–604.

Ramos, V.A., Zapata, T., Cristallini, E., Introcaso, A., 2004. The Andean thrust system-Latitudinal variations in structural styles and orogenic shortening. In: McClay (Ed.). Thrust tectonics and hydrocarbon systems: AAPG Memoir 82, pp. 30–50.

Randall, D.E., 1998. A new Jurasic-Recent apparent polar wander path for South America and a review of central Andean tectonic models. Tectonophysics 299, 49–74.

Rapalini, A.E., 2005. The accretionary history of southern South America from the latest Proterozoic to the Late Palaeozoic: some palaeomagnetic constraints. In: Vaughan, A.P.M., Leat, P.T., Pankhurst, R.J. (Eds.), Terrane Processes at eht Margins of Gondwana. Geological Society of London, Special Publications, 246, pp. 305–328.

Rapalini, A.E., Cingolani, C.A., 2004. First Late Ordovician paleomagnetic pole for the Cuyania (Precordillera) terraine of western Argentina: a microcontinent of a Laurentian plateau? Gondwana Research 7 (4), 1089–1104.

Rapela, C.W., Casquet, C., Baldo, E., Dahlquist, J., Pankhurst, R.J., Galindo, C., Saavedra, J., 2001. Las Orogénesis del Paleozoico Inferior en el margen proto-andino de América del Sur, Sierras Pampeanas, Argentina. Journal of Iberian Geology 27, 23–41.

Rapela, C.W., Pankhurst, R.J., Casquet, C., Baldo, E., Saavedra, J., Galindo, C., Fanning, C.M., 1998. The Pampean Orogeny of the suthern proto-Andes: Cambrian continental collision in the Sierrras de Córdoba. In: Pankhurst, Rapela (Eds.), The Proto-Andean Margin of Gondwana. Geological Society, Special Publication 142, pp. 181–217.

Richards, D.R., Butler, R.F., 2004. Vertical-axis rotations determined from paleomagnetism of Mesozoic and Cenozoic strata of the Bolivian Andes. Journal of Geophysical Research 109 (B07104), 21.

Rubiolo, D., 2001. Descripción geológica de la Hoja 2366-II/2166-IV, La Quiaca. Servicio Geológico Minero Argentino e Instituto de Geología y recursos Minerales. Boletín No. 246.

Russo, A., 1954. Levantamiento geológico en los Departamentos Orán, Iruya y Santa Victoria (Pcia de Salta) y Humahuaca (Pcia de Jujuy). Unpublished report for YPF, Bs.As, Argentina.

Russo, A., Barcat, C., 1978. Relación entre los sedimentos cámbricos y ordovícicos en el norte de las provincias de Salta y Jujuy. Acta Geológica Lilloana, Suplemento 14 (2), 117–119.

Sánchez, M.C., Salfity, J.A., 1999. La cuenca cámbrica del Grupo Mesón en el Noroeste Argentino: desarrollo estratigráfico y paleogeográfico. Acta Geológica Hispánica 34 (2–3), 123–139.

Sánchez Bettucci, L., Rapalini, A.E., 2002. Paleomagnetism of the Sierras de las Ánimas Complex, Southern Uruguay: its implications in the assembly of western Gondwana. Precambrian Research 118, 243–265.

Schwartz, J.J., Gromet, L.P., 2004. Provenance of a late Proterozoic-early Cambrian basin, Sierras de Córdoba, Argentina. Precambrian Research 129, 1–21.

Simpson, C., Lan, R.D., Gromet, L.P., Miro, R., Northrup, C.J., 2003. Paleozoic deformation in the Sierras de Córdoba and Sierra de las Minas, eastern Sierras Pampeanas, Argentina. Journal of South American Earth Sciences 15, 749–764.

Somoza, R., Tomlinson, A., 2002. Paleomagnetism in the Precordillera of northern Chile (22°30′S): implications for the history of tectonic rotations in the Central Andes. Earth and Planetary Science Letters 194, 369–381.

Somoza, R., Singer, S., Coira, B., 1996. Paleomagnetism of upper Miocene ignimbrites at Puna: an analysis of vertical-axis rotations in the Central Andes. Journal of Geophysical Research 101 (B5), 11.387–11.400.

Spagnuolo, C.M., 2004. Geología del anticlinal de Matancillas y paleomagnetismo del Grupo Mesón en la provincia de Salta. Department of Geological Sciences, University of Buenos Aires, Graduate thesis, unpublished. 101 pp.

Spagnuolo, C.M., Rapalini, A.E., Astini, R.A., in press. Paleomagnetic confirmation of Paleozoic clockwise rotation of the Famatina Ranges (NW Argentina): implications for the evolution of the SW margin of Gondwana. Geophysical Journal International.

Spagnuolo, C.M., Rapalini, A.E., Astini, R.A., 2005. Preliminary paleomagnetic results from the cambrian Mesón group: implications for the Bolivian Orocline. 6° International Symposium of Andean Geodynamics, Barcelona, España.

Tan, X., Kodama, K.P., 2002. Magnetic anisotropy and paleomagnetic inclination shallowing in red beds: evidence from the Mississippian Mauch Chunk Formation, Pennsylvania. Journal of Geophysical Research 107 (B11), 2311.

Trindade, R.I.F., D'Agrella-Filho, M.S., Epof, I., Brito Neves, B.B., 2006. Paleomagnetism of Early Cambrian Itabaiana mafic dikes (NE Brazil) and the final assembly of Gondwana. Earth and Planetary Science Letters 244, 361–377.

Trindade, R.I.F., D'Agrella-Filho, M.S., Babinsky, M., Font, E., Brito Neves, B.B., 2004. Paleomagnetism and geochronology of the Bebedouro cap carbonate: evidence for continental-scale Cambrian remagnetization in the São Francisco craton, Brazil. Precambrian Research 128 (1–2), 83–103.

Turner, J.C.M., 1960. Estratigrafía de la Sierra de Santa Victoria y adyacencias. Boletín de la Academia Nacional de Ciencias de la República Argentina 41 (2), 163–196.

Turner, J.C.M., Mendez, V., 1975. Geología del sector oriental de los Departamentos de Santa Victoria e Iruya, Provincia de Salta. República Argentina. Boletín de la Academia Nacional de Ciencias, 51, pp. 11–24.